

Spam Detection using Machine Learning

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Introduction:

This project focuses on building a simple spam detection system using basic machine learning techniques. The aim is to classify messages as spam or not spam (ham).

Objective:

The objective is to understand how machine learning can be used for text classification and to implement a simple model suitable for beginners.

Dataset:

A small dataset of messages is used, where each message is labeled as spam or ham.

Methodology:

1. Data preprocessing: cleaning and converting text to lowercase.
2. Feature extraction: using Bag of Words.
3. Model: Naive Bayes classifier.
4. Training and testing the model.

Implementation:

Python and basic libraries such as sklearn are used. CountVectorizer converts text into numerical form and MultinomialNB is used for classification.

Result:

The model is able to classify messages with reasonable accuracy for a basic dataset.

Conclusion:

This project helps in understanding the fundamentals of machine learning and its application in spam detection.